

PROJECT 2: LOGISTIC REGRESSION
MASM22/FMSN30/FMSN30F/FMSN40: LINEAR AND LOGISTIC
REGRESSION (WITH DATA GATHERING), 2026
Peer assessment version: **17.00 on Monday 11 May**
Peer assessment comments: **17.00 on Tuesday 12 May**
Final version: **17.00 on Wednesday 13 May**

Introduction

As we found out in Project 1, there were a few municipalities that had a large influence on the linear regression model, suggesting that we should not use them when fitting the model for the log-PM₁₀ particle emissions.

In this project we will use all the original data, including the problematic ones, and instead model the probability of a municipality having an unusually high (in the top third) particle emission rate using binary logistic regression.

The data is still located in `Project1data.xlsx`.

Part 1. Introduction to logistic regression

- Lec.7 1(a). Define a high emission rate as a PM₁₀-value lying in the upper tertile (upper third) of the data. Create a new variable, `pm10_high`, with value 1 if PM₁₀ is in the upper tertile and 0 otherwise. *Hint:* use `quantile(PM10, probs = )` to find the cut-off value.
- In order to make it easier to colour the observations according to the values of this response variable, as well as for model validation later, it will be convenient to also create a separate factor version. Add both new variables to the data set. You can now plot with, e.g. `aes(..., y = pm10_high)` and get 0 and 1 on the y-axis, while `aes(..., color = pm10_cat)` will give you different colours for "High" and "Low". You can use either version as dependent variable in a logistic regression since R uses the last category (1 or "High") as "success".
- Report the cut-off value and present a frequency table of the number of municipalities with Low and High emissions, respectively.
- Lec.7 1(b). Turn `Part` into a factor variable (again), using "Göteborg" as reference category.
- Examine the relationship between a high emission level and whether the municipality is located in Göteborg, Svealand or Norrland by counting the number of observations in each of the six combinations.
- Use these raw numbers (not a regression model) to estimate the probability p and the corresponding odds, $p/(1 - p)$, and log odds, of having a high emission level for each of the three Parts. Also calculate the odds ratios for Svealand and Norrland using Göteborg as reference category, and the corresponding log odds ratios. Present the results in a table (Table 1(b)).
- Describe how the odds of having a high emission rate changes when we change from Göteborg to Norrland.
- Lec.8 1(c). Now fit a logistic regression model, *Model 1(c)*, with `Part` as explanatory variable. Present a table with the β -estimates, their standard errors, their 95 % profile likelihood confidence intervals, the corresponding e^β , and their 95 % confidence intervals.

Identify the odds and odds ratios in Table 1(b) that are connected to the different e^β from the model.

Use *Model.1(c)* to estimate the log-odds of high emissions for Götaland, Svealand and Norrland, together with their standard errors and 95 % confidence intervals. Also calculate the corresponding probabilities and 95 % confidence intervals.

Use a suitable test to determine whether there are any significant differences in the probability of a high level of emissions between the three parts of Sweden. Report the type of test you use, the null hypothesis, the value of the test statistic, its distribution, the P-value and the conclusion.

Lec.8 1(d). Ignore Part and turn back to the fuel consumption instead.

Plot the 0/1 variable `pm10_high` against `Fuel` and add a moving average with `geom_smooth()`. Also plot a version using `log-Fuel` instead.

Comment on the two plots. Should we still use the log-transformed `Fuel`, as in Project 1, and does it seem reasonable to use (log) `Fuel` consumption as an explanatory variable?

Fit a simple logistic regression, *Model.1(d)*, using `log(Fuel)` as explanatory variable. Report the β -estimates with 95 % confidence intervals, as well as the e^β -estimates and their confidence intervals.

Add the estimated probability of a high emission rate, and its confidence interval, to the plot.

Use a suitable test to determine if there is a significant relationship between a high emission rate and the fuel consumption in a municipality. State the null hypothesis H_0 , what type of test you use, and why you choose that type, the value of the test statistic, the distribution of the test statistic when H_0 is true, the P-value and the conclusion.

How does the `Fuel` consumption change when the log-fuel consumption increases by one unit? How does the odds of having a high emission rate change when the log-fuel consumption increases by 1 unit, according to the model? How does the odds change when the fuel consumption increases by 10 %? Also present 95 % confidence intervals for these changes.

Lec.8 1(e). Calculate the leverage values for *Model.1(d)* and plot them against `log-Fuel`, against the linear predictor $x_i\hat{\beta}$, and against the predicted probabilities $\hat{p}_i$. Add horizontal reference lines at the minimal value $1/n$ and at $2(p+1)/n$ and make sure the y-axis includes zero.

Relate the general behaviour of the leverage to the behaviour of the estimated probabilities. Why are the two "bumps" in the leverage located where they are? *Hint*: Where does the S-curve change its slope?

Lec.8 1(f). Calculate McFadden's adjusted pseudo R^2 , AIC and BIC for *Model.1(c)* and *Model.1(d)* and decide whether part of Sweden or fuel consumption seems more important for explaining the differences in the probability of having a high emission rate in a municipality.

Part 2. Multiple logistic regression

Coming soon.

Part 3. Goodness-of-fit

Coming soon.